

Bridging the Data Gap: A Human-Validated Pipeline for Culturally Aligned Topic Detection in Black Media

Anonymous submission

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Abstract

In the age of generative artificial intelligence, equitable AI development requires training data that reflects the full breadth of human experience, yet Black media publications remain systematically absent from major news corpora such as C4 and Common Crawl. This absence embeds structural blind spots into language models and limits their cultural alignment for Black communities. We present a modular, open-source pipeline that continuously scrapes, normalizes, and thematically annotates articles from ten prominent Black media publications on a rolling monthly horizon, delivering approximately 1,200 articles per cycle tagged against a curated taxonomy of twelve social-conflict topics. To move beyond pipeline description, we construct a 400-article human-annotated benchmark across two independent annotators and conduct a rigorous comparative evaluation of six tagging approaches: keyword matching, semantic similarity, three hybrid variants, and a supervised classifier trained on sentence embeddings. We establish a human agreement baseline through expert re-labeling and inter-annotator agreement analysis, finding macro Cohen’s kappa of 0.33 to 0.38, then show that our best method—a per-topic logistic regression combining embeddings with keyword and semantic signals—reaches 77 to 91 percent of that baseline with a macro F1 of 0.322 and kappa of 0.297 under a leak-free cross-annotator validation protocol. We release the pipeline code, taxonomy, and annotation metadata to support replication and extension to other under-represented media ecosystems. This work demonstrates that rigorous, human-validated topic tagging of community media is achievable and offers a transferable recipe for low-resource, equity-focused dataset infrastructure.

1 Introduction

Large language models inherit the contours of their training data. When a community’s press is missing from the corpora these models are trained on, the community’s framing of its own issues—how policing, housing, maternal health, or economic equity are discussed from

within—is missing too. The result is a structural blind spot: models that can fluently summarize mainstream coverage of a police-reform debate but cannot reproduce, or even recognize, the perspective of the Black press that has covered such debates continuously for two centuries Washburn (2006).

This gap is measurable, not merely rhetorical. C4, the corpus underlying the T5 family of models Raffel et al. (2020), was constructed from the April 2019 Common Crawl snapshot. Auditing that snapshot, we find that the ten Black media publications monitored in this work account for a combined total of fewer than 8,000 captured pages—several orders of magnitude below single mainstream outlets, with nytimes.com alone ranking among the largest domains in C4 Dodge et al. (2021). One of our publications, Capital B News, was founded in 2022 and therefore cannot appear in C4 or in any model trained on it. Section 4 details this audit.

Prior infrastructure work, including our own, has responded by building collection pipelines for community media. But collection alone leaves a critical question unanswered, one that reviewers of such systems rightly press: are the produced topic labels reliable enough for downstream research use? A pipeline that tags articles with unvalidated heuristics produces a dataset whose labels cannot be trusted, and an untrustworthy dataset serves the community it documents poorly.

This paper answers that question for our pipeline, and in doing so offers a template for answering it in other low-resource media ecosystems. We make four contributions:

1. An extended collection pipeline. We expand coverage from seven to ten Black media publications, scraped on a rolling monthly horizon (~1,200 articles per cycle), normalized to a shared schema, and tagged against a twelve-topic social-conflict taxonomy, fully automated via CI/CD (Section 3).
2. A human-annotated benchmark with an agreement baseline. Two independent annotators labeled 200 articles each across all twelve topics. An expert re-labeling of all 400 articles yields inter-annotator agreement of macro $\kappa = 0.33$ –0.38, establishing the human baseline against which any automated tagger must be read (Section 5).

3. A leak-free comparative evaluation of six tagging methods. We compare keyword matching, semantic similarity, and three hybrid variants under a cross-annotator validation protocol, and we document a threshold-tuning leak in our own earlier evaluation whose correction reduces apparent performance by 0.07 macro F1—a cautionary result for evaluation practice in this space (Section 7).
4. A validated supervised classifier. A per-topic logistic regression over sentence embeddings, keyword, and semantic signals achieves macro F1 0.322 and κ 0.297 under the leak-free protocol—77–91% of the human agreement baseline—and is adopted as the pipeline’s production tagger (Sections 6–7).

We frame this system as a foundational, incrementally validated pipeline rather than a finished large-scale resource. Throughout, we prioritize transparency about data sparsity, annotation difficulty, and the limits of our metrics over scale or performance claims (Section 8).

2 Related Work

Data equity and corpus documentation. A growing literature documents whose voices web-scale corpora contain. Dodge et al. (2021) audit C4 and show its domain distribution is dominated by a small set of mainstream outlets; Bender et al. (2021) argue that undocumented, skewed corpora encode hegemonic viewpoints; Jo and Gebru (2020) draw on archival practice to propose deliberate, consent-aware collection of socio-cultural data; and Gebru et al. (2021) formalize dataset documentation itself. Disparities first quantified in vision systems Buolamwini and Gebru (2018) recur in text: under-representation at the corpus level becomes mis-representation at the model level. Our work operationalizes these critiques for one concrete ecosystem—the Black press—by building and validating the missing data infrastructure.

The Black press. The Black press has served as a parallel public sphere since Freedom’s Journal (1827), covering issues mainstream outlets ignored or distorted Washburn (2006). Its contemporary digital successors—from legacy brands to digital natives founded in the last decade—are precisely the outlets our audit shows to be absent from mainstream corpora, motivating focused collection.

Topic detection and annotation quality. Topic assignment methods span keyword and seed-phrase matching, embedding-based semantic similarity Reimers and Gurevych (2019), and supervised classification Pedregosa et al. (2011). Orthogonal to method choice is label quality: chance-corrected agreement Cohen (1960)

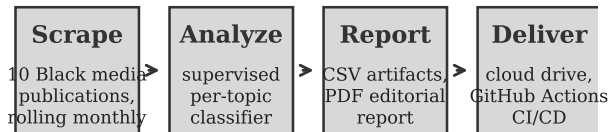


Figure 1: The four-stage pipeline. This paper’s contribution is the validation of the Analyze stage: the supervised classifier adopted there is the best of six methods evaluated against a 400-article human benchmark.

and its interpretation Landis and Koch (1977); Artstein and Poesio (2008) are standard in computational linguistics but remain rare in dataset-infrastructure papers, where system throughput is more commonly reported than label validity. We treat agreement analysis as a first-class component of the pipeline.

3 Pipeline Overview

The pipeline (Figure 1) runs monthly via GitHub Actions and comprises four stages. **Scrape**: source-specific scrapers (WordPress REST, HTML, RSS fallback) collect every article published by ten Black media publications within a rolling window matching the calendar month, yielding $\sim 1,200$ articles per cycle. **Analyze**: each article—title, description, and body text—is tagged against the twelve-topic social-conflict taxonomy of Table 1 by the production tagger (the supervised classifier validated in this paper). **Report**: the run exports research-ready CSV artifacts and a styled PDF editorial summary. **Deliver**: artifacts are deposited to a shared cloud drive for non-technical stakeholders.

The taxonomy was curated with domain collaborators to cover the social-conflict issues most consistently central to Black media coverage. Articles may carry multiple topics or none; in practice a large share of the feed (sports, entertainment, lifestyle) falls outside the taxonomy entirely, which has direct consequences for annotation (Section 5).

4 Measuring the Data Gap

Earlier presentations of this pipeline asserted that its sources are missing from mainstream corpora; here we measure it. C4 Raffel et al. (2020) was filtered from the Common Crawl snapshot CC-MAIN-2019-18. We queried the Common Crawl index for every capture of each of our ten publication domains in that snapshot (Table 2). Because C4’s filters (English-only, deduplication, blocklists, length heuristics) only remove pages,

Topic	Human positives	
	Set 1	Set 3
Policing / Public Safety	23	13
Economic Equity / Wealth Gap	47	30
Housing / Displacement	14	9
Environmental Justice	14	7
Voter Suppression	13	3
Criminal Justice Reform	7	8
School Funding	7	3
Redlining / Fair Housing	5	0
Reparations	5	0
Book Bans / Anti-DEI	4	4
Maternal Health	4	4
Anti-Black Surveillance	3	0
Total label decisions	$400 \times 12 = 4,800$	

Table 1: The twelve-topic social-conflict taxonomy with the number of positive human labels in each 200-article annotation set. The long tail is severe: three topics have zero positives in Set 3, a sparsity that shapes every result in this paper.

these counts upper-bound each domain’s possible C4 presence.

The combined measured footprint of nine of our ten domains—7,766 raw captures before C4’s aggressive filtering—is several orders of magnitude smaller than that of a single mainstream outlet. The Capital B result generalizes: any publication founded after a corpus’s snapshot date is structurally invisible to models trained on it, and community media skews young. Continuous collection, not one-time archival, is therefore the appropriate remedy, which is what the pipeline provides.

5 Human Annotation and Agreement

Protocol. Two independent annotators each labeled a disjoint set of 200 articles sampled from a pipeline run, marking, for each article, every applicable topic among the twelve (binary decisions; multi-label; no topic is also valid). This yields 4,800 label decisions. To measure agreement, a domain expert on the research team independently re-labeled all 400 articles under the same guidelines.

Results. Figure 2 reports per-topic Cohen’s κ Cohen (1960) between the expert and each original annotator. Macro-averaged over topics, agreement is $\kappa = 0.327$ against Annotator 1 and $\kappa = 0.384$ against Annotator 3; pooled over all decisions, $\kappa = 0.410$ and 0.531 respectively—fair-to-moderate on conventional scales Landis and Koch (1977).

Domain	Captures in CC-MAIN-2019-18
blavity.com	3,774
theroot.com	1,291
ebony.com	911
thegrio.com	881
essence.com	407
newsone.com	203
afrotech.com	186
21ninety.com	113
capitalbnews.org	0
travelnoire.com	n/a*
Combined (measured)	7,766

Table 2: Presence of our ten publication domains in the Common Crawl snapshot from which C4 was built. *Index query unresolved at submission time. For scale, nytimes.com is among the largest domains in all of C4 Dodge et al. (2021). Capital B News (founded 2022) post-dates the snapshot entirely.

Why agreement is moderate—and why we report it anyway. We treat the manual annotation as a baseline standard, not perfect ground truth, and the κ values as a finding about the task rather than a defect of the study. Three factors depress agreement, all visible in Table 1 and Figure 2. First, topic sparsity: five topics have five or fewer positives per set, and three have none at all in Set 3; with so few positives, a single disagreement swings κ dramatically, and κ is undefined or uninformative at zero prevalence Artstein and Poesio (2008). Second, feed composition: much of the scraped stream is sports, entertainment, and lifestyle coverage outside the taxonomy, so annotators spend most decisions confirming absence, and the rare positive cases concentrate the ambiguity. Third, boundary overlap: the hardest topics are those with genuinely contested boundaries—Economic Equity absorbs stories that also touch housing or schooling, and platform-side miscategorization (e.g., policing stories filed under generic “politics”) seeds anchoring disagreements. An automated tagger evaluated against labels of this character cannot meaningfully exceed them; it can only approach them. That is the yardstick we adopt.

6 Tagging Methods

We evaluate six methods, in increasing order of supervision.

Keyword tagger. Curated seed phrases per topic, matched against title, description, and body. High precision on explicit mentions; blind to paraphrase.

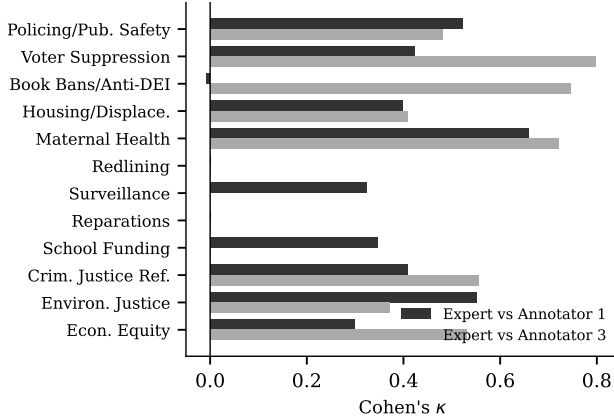


Figure 2: Per-topic inter-annotator agreement (Cohen’s κ) between the expert re-labeling and each original annotator. Zero bars are topics with no positive labels from either party; agreement there is undefined in practice.

Semantic tagger. Cosine similarity between the article embedding and per-topic seed-text embeddings, using all-MiniLM-L6-v2 Reimers and Gurevych (2019), with a global decision threshold. Recovers paraphrase; noisier near the threshold.

Hybrid OR / AND. Union and intersection of the two base taggers’ positive sets. OR maximizes recall at heavy precision cost; AND the reverse.

Hybrid Soft Boost. Keyword matches always fire; additionally, the semantic score can fire a topic if it clears a per-topic threshold grid-searched to maximize F1. We evaluate this in two forms. The in-sample form tunes thresholds on the same annotator set used for scoring—this is the version with a tuning leak, retained deliberately as an upper bound and as a documented cautionary result. The cross-validated (CV) form tunes thresholds on one annotator’s 200 articles and applies them to the other’s, eliminating the leak.

Supervised classifier. One logistic regression per topic Pedregosa et al. (2011), with class-balanced weighting, over a feature vector concatenating (i) the 384-dimensional sentence embedding of the article text, (ii) the keyword tagger’s binary decision, and (iii) the raw semantic score for that topic. Training and testing follow the same cross-annotator protocol as Soft Boost (CV): train on Annotator 1’s set, test on Annotator 3’s, and vice versa. Topics with fewer than two positive training examples fall back to the keyword decision; this occurred for three topics in one direction (Redlining, Surveillance, Reparations—exactly the zero-prevalence topics of Table 1) and is flagged in all reported results.

Method	Macro F1	Macro κ
Keyword	0.214	0.183
Semantic	0.266	0.225
Hybrid OR	0.229	0.181
Hybrid AND	0.259	0.244
Hybrid Soft Boost (in-sample) [†]	0.336	0.297
Hybrid Soft Boost (CV)	0.267	0.224
Supervised (CV)	0.322	0.297
Human baseline (IAA)	—	0.327–0.384

Table 3: Pooled macro-averaged results over both annotator sets. [†]In-sample Soft Boost tunes thresholds on the evaluation data and is reported only as an upper bound; all other tuned or trained methods use leak-free cross-annotator validation. The supervised classifier is the best honest method and matches the leaked upper bound’s κ .

7 Evaluation

Protocol. All methods are scored against both annotator sets on per-topic precision, recall, F1, and Cohen’s κ , macro-averaged over the twelve topics and pooled across the two sets. Methods requiring tuning or training never see the set they are scored on: this cross-annotator validation is a deliberately harsh test, since the annotators themselves agree only at $\kappa = 0.33$ – 0.38 , so any tuned artifact must survive both sampling variance and genuine annotator shift.

The threshold-tuning leak. Table 3 reports Soft Boost both ways. In-sample tuning yields macro F1 0.336; the leak-free CV form yields 0.267. The 0.07 gap—21% of the apparent score—was pure tuning optimism, and the honest hybrid is statistically indistinguishable from the untuned semantic baseline (0.266). We highlight this because per-topic threshold tuning on evaluation data is common, tempting, and in low-prevalence regimes like ours, badly misleading.

Main result. The supervised classifier attains macro F1 0.322 and κ 0.297 under the leak-free protocol (Figure 3)—a 21% relative F1 improvement over the best honest alternative, and a κ equal to the leaked Soft Boost upper bound. Measured against the human agreement baseline, the classifier reaches 91% of the Set 1 ceiling (0.297/0.327) and 77% of the Set 3 ceiling (0.297/0.384). Both cross-annotator directions agree closely (F1 0.326 testing on Set 3; 0.319 testing on Set 1), indicating the result is not an artifact of one annotator’s style.

Per-topic behavior. Figure 4 breaks the supervised result down by topic and direction. Performance tracks prevalence: the well-populated topics (Economic Equity, Policing, Housing, Criminal Justice Reform, Environmental Justice) reach F1 0.42–0.62, while the zero-

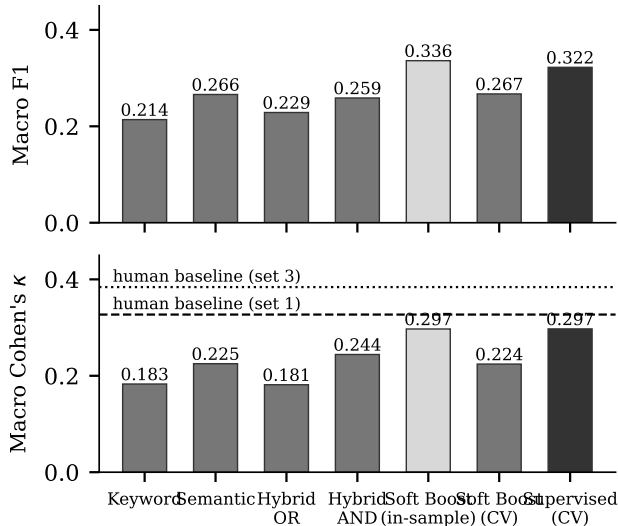


Figure 3: Pooled macro F1 (top) and Cohen’s κ (bottom) for all methods. Dashed and dotted lines mark the human agreement baselines from the IAA study. The supervised classifier (dark) is the only honest method whose κ approaches the human range; the in-sample Soft Boost bar (pale) is the documented leaked upper bound.

and near-zero prevalence topics (Redlining, Reparations, Surveillance) are unmeasurable—no method can be evaluated on topics for which humans found no positive examples. We report these as unmeasurable rather than as zeros of model quality; expanding annotation for rare topics is future work, not a solved problem.

Production adoption. On these results the supervised classifier, retrained on all 400 labeled articles, replaces the keyword matcher as the pipeline’s production tagger. The frozen 400-article benchmark and evaluation scripts run in CI, so future modifications to the tagger or taxonomy are automatically scored against the human baseline before deployment.

8 Limitations and Discussion

Moderate agreement, honestly reported. Our human baseline (κ 0.33–0.38 macro) is fair-to-moderate, and we have argued this reflects task structure—sparsity, feed composition, boundary overlap—rather than careless annotation. Nonetheless, all downstream numbers inherit this uncertainty, and readers should treat per-topic scores on sparse topics as indicative at best. Scale versus validity. Restricting collection to articles plainly within the twelve topics would raise every metric in this paper, at the cost of a smaller, less representative resource. For this submission we kept the full-feed scale and report the resulting sparsity transparently; stricter

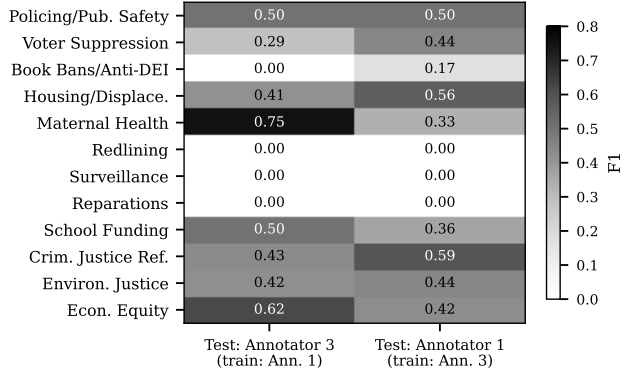


Figure 4: Per-topic supervised F1 in both cross-annotator directions. Performance concentrates where human labels exist (cf. Table 1); blank-prevalence topics are unmeasurable.

subtopic filtering is a tunable choice we leave to dataset users. Anchoring risk. Original annotation sheets displayed the keyword pipeline’s predictions, which may anchor annotators toward them; the expert re-labeling was performed blind to predictions, but the original labels were not. Two annotators, one expert. The IAA study spans two annotator–expert pairs; a larger annotator pool would tighten the baseline estimate. No downstream experiment yet. We validate label quality, not downstream impact on model behavior; demonstrating that this data measurably improves cultural alignment of a language model is the natural next study, enabled but not performed by this work.

9 Conclusion and Future Work

We presented a continuously running pipeline for collecting and thematically tagging Black media coverage, and—the core contribution—a human-validated evaluation of its tagging stage. A measured corpus audit grounds the motivating data gap; a 400-article benchmark with expert re-labeling establishes a human agreement baseline; a documented threshold-tuning leak and its correction establish honest evaluation practice; and a supervised classifier reaching 77–91% of the human baseline becomes the production tagger. The system is foundational by design: future work includes broader annotation for rare topics, stricter subtopic filtering as a user-facing option, extension to additional community media ecosystems, and downstream experiments measuring whether culturally grounded data of this kind improves the alignment of generative models with the communities it represents. Code, taxonomy, evaluation scripts, and annotation metadata (URLs, labels, and scores; not full article text, respecting publisher copyright) will be released upon publication.

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